C

IB MATH: APPLICATIONS & INTERPRETATION • UNIT 6.01

Linear Regression & OLS

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

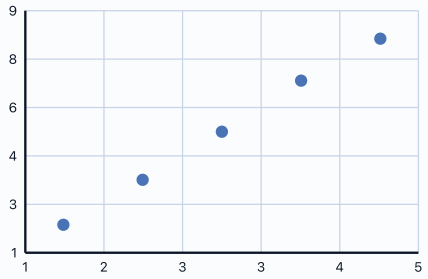
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 A
Q6 A	Q7 A	Q8 B	Q9 B	Q10 A
Q11 C	Q12 B	Q13 A	Q14 A	Q15 C
Q16 D	Q17 C	Q18 C	Q19 D	Q20 C
Q21 C	Q22 C	Q23 A	Q24 B	Q25 A

Linear Regression & OLS Practice Exercises

1 MCQ

A fitted simple linear regression model is $\hat{y} = 7x + 18$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?



- **A** 46 ✓
- **B** 49
- **C** 42
- **D** 28

WORKED SOLUTION & VERIFICATION

Substitute $x = 4$ into the regression equation: $\hat{y} = 7(4) + 18 = 28 + 18 = 46$.

2 MCQ

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 20$ and the Total Sum of Squares is $\text{SST} = 100$. What is the coefficient of determination R^2?

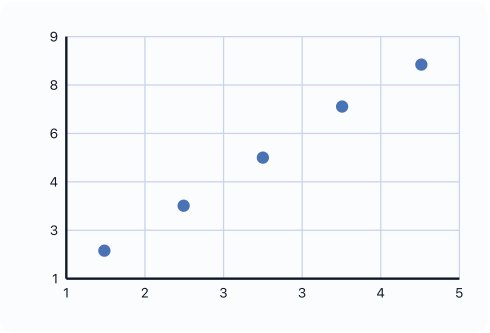
- **A** $R^2 = 0.80$ ✓
- **B** $R^2 = 0.20$
- **C** $R^2 = 0.90$
- **D** $R^2 = 0.25$

WORKED SOLUTION & VERIFICATION

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{20}{100} = 1 - 0.20 = 0.80$.

3 MCQ

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 40$ and the Total Sum of Squares is $\text{SST} = 200$. What is the coefficient of determination R^2?



- **A** $R^2 = 0.80$ ✓
- **B** $R^2 = 0.20$
- **C** $R^2 = 0.90$
- **D** $R^2 = 0.25$

WORKED SOLUTION & VERIFICATION

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{40}{200} = 1 - 0.20 = 0.80$.

4 MCQ

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 60$ and the Total Sum of Squares is $\text{SST} = 300$. What is the coefficient of determination R^2?

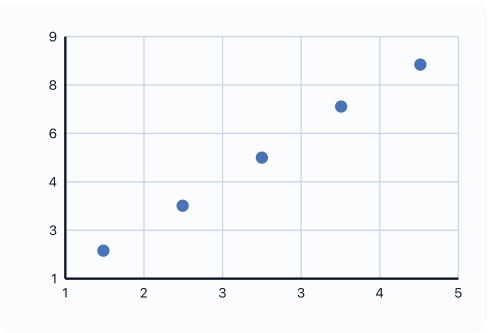
- **A** $R^2 = 0.80$ ✓
- **B** $R^2 = 0.20$
- **C** $R^2 = 0.90$
- **D** $R^2 = 0.25$

WORKED SOLUTION & VERIFICATION

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{60}{300} = 1 - 0.20 = 0.80$.

5 MCQ

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 80$ and the Total Sum of Squares is $\text{SST} = 400$. What is the coefficient of determination R^2?



- **A** $R^2 = 0.80$ ✓
- **B** $R^2 = 0.20$
- **C** $R^2 = 0.90$
- **D** $R^2 = 0.25$

WORKED SOLUTION & VERIFICATION

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{80}{400} = 1 - 0.20 = 0.80$.

6 MCQ

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 100$ and the Total Sum of Squares is $\text{SST} = 500$. What is the coefficient of determination R^2?

- **A** $R^2 = 0.80$ ✓
- **B** $R^2 = 0.20$
- **C** $R^2 = 0.90$
- **D** $R^2 = 0.25$

WORKED SOLUTION & VERIFICATION

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{100}{500} = 1 - 0.20 = 0.80$.

7 MCQ

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 120$ and the Total Sum of Squares is $\text{SST} = 600$. What is the coefficient of determination R^2?

- **A** $R^2 = 0.80$ ✓
- **B** $R^2 = 0.20$
- **C** $R^2 = 0.90$
- **D** $R^2 = 0.25$

WORKED SOLUTION & VERIFICATION

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{120}{600} = 1 - 0.20 = 0.80$.

8 MCQ

For a bivariate feature dataset, $\text{Cov}(x, y) = 12$, $\text{Var}(x) = 4$, $\bar{x} = 5$, and $\bar{y} = 25$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

- A $\hat{\beta}_1 = 0.3333333333333333,\ \hat{\beta}_0 = 10$
- B $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 10$ ✓
- C $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 25$
- D $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 5$

WORKED SOLUTION & VERIFICATION

The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{12}{4} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 25 - (3)(5) = 25 - 15 = 10$.

9 MCQ

Consider a dataset of n observations (x_i, y_i) for $i = 1, \ldots, n$, where x_i are scalar predictor variables and y_i are scalar response variables. The ordinary least squares (OLS) estimates for the intercept β_0 and the slope β_1 of the linear model $y = \beta_0 + \beta_1 x + \epsilon$ are given by $\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Suppose we have a dataset where $n = 5$, $\sum x_i = 15$, $\sum y_i = 30$, $\sum x_i^2 = 55$, $\sum y_i^2 = 210$, and $\sum x_i y_i = 105$. A new data point (x_6, y_6) is added to the dataset. If the new regression line that minimizes the sum of squared errors for the $n + 1 = 6$ points is parallel to the original regression line (i.e., $\hat{\beta}_1^{\text{new}} = \hat{\beta}_1^{\text{old}}$), what is the value of y_6 in terms of x_6?

- A $y_6 = 1.5x_6 + 0.5$
- B $y_6 = 1.5x_6 + 1.5$ ✓
- C $y_6 = 2x_6 + 0$
- D $y_6 = 2x_6 + 1.5$

WORKED SOLUTION & VERIFICATION

For the original $n = 5$ observations, $\bar{x} = \frac{15}{5} = 3$ and $\bar{y} = \frac{30}{5} = 6$. The sums of squares are $S_{xx} = 55 - 5(3^2) = 10$ and $S_{xy} = 105 - 5(3)(6) = 15$, giving slope $\hat{\beta}_1 = \frac{15}{10} = 1.5$ and intercept $\hat{\beta}_0 = 6 - 1.5(3) = 1.5$. When a 6th point (x_6, y_6) lying on the existing regression line $y_6 = 1.5x_6 + 1.5$ is added, the residual is zero and the OLS slope remains unchanged at 1.5.

10 MCQ

Considering a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 40$ and the Total Sum of Squares is $\text{SST} = 200$. What is the coefficient of determination R^2?

- A $R^2 = 0.80$ ✓
- B $R^2 = 0.25$
- C $R^2 = 0.20$
- D $R^2 = 0.90$

WORKED SOLUTION & VERIFICATION

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{40}{200} = 1 - 0.20 = 0.80$.

11 MCQ

Examining a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 60$ and the Total Sum of Squares is $\text{SST} = 300$. What is the coefficient of determination R^2?

A $R^2 = 0.20$

B $R^2 = 0.90$

C $R^2 = 0.80$ ✓

D $R^2 = 0.25$

WORKED SOLUTION & VERIFICATION

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{60}{300} = 1 - 0.20 = 0.80$.

12 MCQ

For the configuration in a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 80$ and the Total Sum of Squares is $\text{SST} = 400$. What is the coefficient of determination R^2?

A $R^2 = 0.20$

B $R^2 = 0.80$ ✓

C $R^2 = 0.90$

D $R^2 = 0.25$

WORKED SOLUTION & VERIFICATION

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{80}{400} = 1 - 0.20 = 0.80$.

13 MCQ

Analyzing a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 100$ and the Total Sum of Squares is $\text{SST} = 500$. What is the coefficient of determination R^2?

A $R^2 = 0.80$ ✓

B $R^2 = 0.90$

C $R^2 = 0.25$

D $R^2 = 0.20$

WORKED SOLUTION & VERIFICATION

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{100}{500} = 1 - 0.20 = 0.80$.

14 MCQ

Within a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 120$ and the Total Sum of Squares is $\text{SST} = 600$. What is the coefficient of determination R^2?

A $R^2 = 0.80$ ✓

B $R^2 = 0.20$

C $R^2 = 0.90$

D $R^2 = 0.25$

WORKED SOLUTION & VERIFICATION

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{120}{600} = 1 - 0.20 = 0.80$.

15 MCQ

For a bivariate feature dataset, $\text{Cov}(x, y) = 24$, $\text{Var}(x) = 8$, $\bar{x} = 5$, and $\bar{y} = 30$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

A $\hat{\beta}_1 = \frac{1}{3},\ \hat{\beta}_0 = 15$

B $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 30$

C $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 15$ ✓

D $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 10$

WORKED SOLUTION & VERIFICATION

The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{24}{8} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 30 - (3)(5) = 30 - 15 = 15$.

16 MCQ

For a bivariate feature dataset, $\text{Cov}(x, y) = 36$, $\text{Var}(x) = 12$, $\bar{x} = 5$, and $\bar{y} = 35$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

A $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 35$

B $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 15$

C $\hat{\beta}_1 = \frac{1}{3},\ \hat{\beta}_0 = 20$

D $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 20$ ✓

WORKED SOLUTION & VERIFICATION

The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{36}{12} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 35 - (3)(5) = 35 - 15 = 20$.

17 MCQ

For a bivariate feature dataset, $\text{Cov}(x, y) = 48$, $\text{Var}(x) = 16$, $\bar{x} = 5$, and $\bar{y} = 40$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

A $\hat{\beta}_1 = \frac{1}{3},\ \hat{\beta}_0 = 25$

B $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 40$

C $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 25$ ✓

D $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 20$

WORKED SOLUTION & VERIFICATION

The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{48}{16} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 40 - (3)(5) = 40 - 15 = 25$.

18 MCQ

For a bivariate feature dataset, $\text{Cov}(x, y) = 60$, $\text{Var}(x) = 20$, $\bar{x} = 5$, and $\bar{y} = 45$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

A $\hat{\beta}_1 = \frac{1}{3},\ \hat{\beta}_0 = 30$

B $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 45$

C $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 30$ ✓

D $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 25$

WORKED SOLUTION & VERIFICATION

The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{60}{20} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 45 - (3)(5) = 45 - 15 = 30$.

19 MCQ

For a bivariate feature dataset, $\mathrm{Cov}(x, y) = 72$, $\mathrm{Var}(x) = 24$, $\bar{x} = 5$, and $\bar{y} = 50$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

A $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 50$

B $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 30$

C $\hat{\beta}_1 = \frac{1}{3},\ \hat{\beta}_0 = 35$

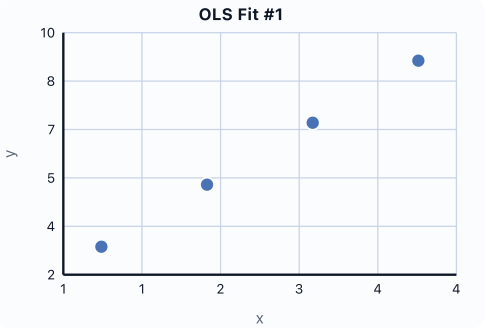
D $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 35$ ✓

WORKED SOLUTION & VERIFICATION

The OLS slope is $\hat{\beta}_1 = \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)} = \frac{72}{24} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 50 - (3)(5) = 50 - 15 = 35$.

20 MCQ

An Ordinary Least Squares regression model (#1) achieves residual sum of squares $\mathrm{SSE} = 5$ on data with total sum of squares $\mathrm{SST} = 20$. Compute the coefficient of determination R^2.



A $\frac{1}{2}$

B $\frac{1}{4}$

C $\frac{3}{4}$ ✓

D 3

WORKED SOLUTION & VERIFICATION

By definition, $R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}} = 1 - \frac{5}{20} = \frac{3}{4}$.

21 MCQ

In dataset #2, sample variance is $s_x^2 = 3$, sample covariance is $s_{xy} = -8$, $\bar{x} = 3$, and $\bar{y} = 10$. Find the OLS slope $\hat{\beta}_1$.

A 14

B $-\frac{3}{8}$

C $-\frac{8}{3}$ ✓

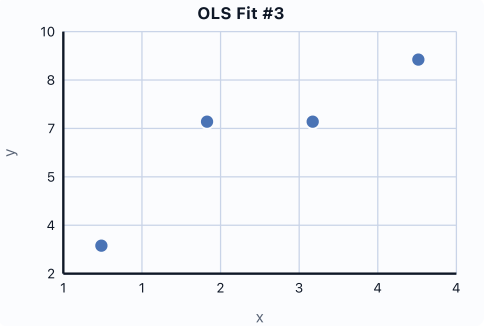
D $\frac{10}{3}$

WORKED SOLUTION & VERIFICATION

The univariate OLS slope estimator is $\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} = \frac{-8}{3} = -\frac{8}{3}$.

22 MCQ

An Ordinary Least Squares regression model (#3) achieves residual sum of squares $\mathrm{SSE} = 15$ on data with total sum of squares $\mathrm{SST} = 70$. Compute the coefficient of determination R^2.



A $\frac{4}{7}$

B $\frac{11}{3}$

C $\frac{11}{14}$ ✓

D $\frac{3}{14}$

WORKED SOLUTION & VERIFICATION

By definition, $R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}} = 1 - \frac{15}{70} = \frac{11}{14}$.

23 MCQ

In dataset #4, sample variance is $s_x^2 = 5$, sample covariance is $s_{xy} = -5$, $\bar{x} = 5$, and $\bar{y} = 10$. Find the OLS slope $\hat{\beta}_1$.

A -1 ✓

B 13

C 25

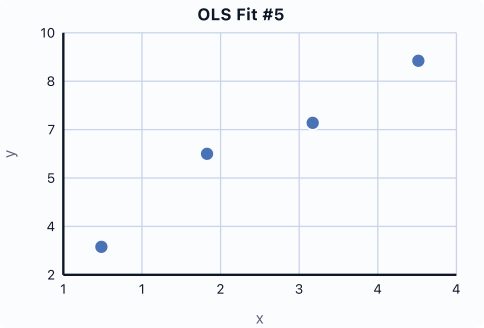
D 2

WORKED SOLUTION & VERIFICATION

The univariate OLS slope estimator is $\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} = \frac{-5}{5} = -1$.

24 MCQ

An Ordinary Least Squares regression model (#5) achieves residual sum of squares $\mathrm{SSE} = 25$ on data with total sum of squares $\mathrm{SST} = 60$. Compute the coefficient of determination R^2.



A $\frac{1}{6}$

B $\frac{7}{12}$ ✓

C $\frac{7}{5}$

D $\frac{5}{12}$

WORKED SOLUTION & VERIFICATION

By definition, $R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}} = 1 - \frac{25}{60} = \frac{7}{12}$.

25 MCQ

In dataset #6, sample variance is $s_x^2 = 7$, sample covariance is $s_{xy} = 9$, $\bar{x} = 2$, and $\bar{y} = 11$. Find the OLS slope $\hat{\beta}_1$.

- **A** $\frac{9}{7}$ ✓
- **B** $\frac{9}{2}$
- **C** $\frac{7}{9}$
- **D** $\frac{11}{2}$

WORKED SOLUTION & VERIFICATION

The univariate OLS slope estimator is $\hat{\beta}_1 = \frac{s_{xy}}{s_x^2} = \frac{9}{7} = \frac{9}{7}$.